

The Neutrino That Weighs Nothing

Why the lightest neutrino is exactly massless — and how the sky and the lab will tell us if we're wrong

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There are three kinds of neutrino, and for a hundred years they have been the shyest particles we know — trillions pass through your thumbnail every second and not one of them touches it. For most of that century we thought they weighed nothing at all. Then, twenty-five years ago, experiments watching them flicker between the three kinds proved they *must* have mass — because a massless thing has no clock, and only a thing with a clock can change. But we still don't know the actual weights. We know two *differences* between the three, measured beautifully. We do not know where the ladder sits.

BST says something sharp about that ladder, and it says it without a single knob to turn: the lightest neutrino weighs exactly nothing. Not “very little.” Zero. And once you accept that one fact, the rest of the neutrino world falls out — the total mass, the ordering, a rare radioactive glow that a detector deep underground is hunting for right now, and a specific answer to a question three enormous experiments will settle this decade. Every one of those is a place BST can be killed.

This is the story of where that zero comes from, what rides on it, and exactly how you'd prove us wrong.

What we claim

Four things, and I'll mark which ones are *ours* — distinctive, the ones we'd stake the theory on — and which are honest but shared with any theory of this shape.

1. The lightest neutrino is exactly massless. (*Distinctive — this is the keystone.*)
2. Neutrinos are their own antiparticles (physicists say *Majorana*), which means a forbidden kind of radioactive decay is not forbidden after all. (*Distinctive.*)
3. The three masses climb in the “normal” order (two light, one heavier), and their total is about 58 thousandths of an electron-volt — a hair's breadth. (*The ordering we predict; the total number, 58, is the honest floor* any normal-order theory gives — we flag it as shared, not a BST fingerprint.)**
4. The rare decay, if it happens, glows at a very specific dimness — between about 1.5 and 3.7 (in the physicists' units, “milli-electron-volts”), and even allowing for the wiggle room in the measured inputs it stays under about 4 — too faint for even the next generation of detectors to see. (*Distinctive, and it has teeth: a glow as bright as the other* mass-ordering would produce — roughly 15 and up — kills us outright, no argument.)**

Zero free parameters. So one clean violation — any one — ends the theory. That's the deal, and it's the point.

How it works

Two ideas, and they turn out to be the *same* idea wearing two coats.

Why “their own antiparticle.” A neutrino spins, and it can spin one of two handednesses — left or right, like a screw. The Standard Model has only ever seen *left*-handed neutrinos; the right-handed partner has never shown up. In most theories that's a mystery you paper over. In BST it isn't a mystery — the shape simply has no room for

a right-handed neutrino. It's not that we haven't found it; the geometry forbids it as a state, the way a chessboard forbids a square that's both black and white. And here's the consequence: if a neutrino has no right-handed partner to pair with, the only way it can carry mass is to pair *with itself* — which is exactly what “being your own antiparticle” means. So Majorana isn't an assumption we bolt on. It's forced by an absence. (*This is our sharpest kind of prediction — one of the “Five Absences” BST stakes itself on: certain things the shape simply cannot contain.*)

Why one of them is massless. Now the beautiful part. The little machine that hands out neutrino masses is built from the geometry, and the geometry has a number at its heart called its *rank*, and that rank is two. A rank-two machine can lift exactly two things. But there are *three* neutrinos. Two get lifted; the third has nowhere to be lifted *from* — it falls into the gap and stays at exactly zero. Not fine-tuned, not arranged. Left over. The way, if you have two hands and three children who each want to be carried, one child walks.

And this is where the two coats turn out to be one coat. This same week, the team credentialed the fundamental operator of BST — the thing that plays the role of “the equation of everything” for this geometry — and found it has exactly one ground state, one lowest note, for the identical reason: rank two leaves exactly one thing unpaired. *One massless neutrino and one ground state are the same fact, seen in two rooms of the house.* Casey has a phrase for this — “the breaking is the content.” The interesting physics is never in the parts that match up neatly; it's in the one thing left over when they don't. Here, the leftover is the massless neutrino.

What it predicts, and what's distinctive versus shared

From “the lightest is zero” plus the two measured differences (which we *use* — we don't invent them), the ladder is pinned:

- The total mass is about 58 meV, in the normal order. (*We predict the ordering; the 58 is the generic floor — honest flag: most normal-order theories land near here, so we don't sell the number itself as a BST fingerprint.*)
- The three mixing angles — how strongly the kinds blend — come out as clean small fractions, and all three sit right on top of the measured values (within a fraction of a standard deviation). (*Consistent, and pretty — but here's the honest tier: a simple fraction landing near a measured number is cheap, and some of these angles have more than one clean fraction that fits. So we hold the individual fractions as consistent, not forced. The one we'd actually bet on is below.*)
- The middle-heavy angle points “upper.” There's a long-standing question of which of two values one particular angle takes — physicists call it the “octant.” BST predicts the upper one, cleanly. (*Distinctive — and falsifiable: the experiments are built to settle this decisively, and a firm lower-octant result would land many standard deviations away from where BST sits. No wiggle.*)

And the piece that is *most* ours, because most theories can hide from it and BST can't:

- The mass total is welded to dark energy. Separately, BST predicts that dark energy does *not* do a certain exotic thing (it doesn't “cross the phantom line”). Most theories with a massless-lightest neutrino can duck a tightening mass measurement by invoking exotic dark energy — it loosens the bound. BST has already forbidden that exotic behavior, so we can't duck. The tight bound applies to us, and our 58 meV sits just 6 meV under its own ceiling. If the cosmic mass total is measured *below* about 58 meV, BST is finished — and we have nowhere to run. (*This coupling is the distinctive exposure: two predictions, one dataset, and we chose the corner with no exit.*)

How to confirm it — three experiments, this decade

We mean the falsifiers literally, so here they are with names and dates:

- The rare glow (neutrinoless double-beta decay). Detectors called LEGEND-1000 and nEXO are hunting a radioactive decay that can *only* happen if neutrinos are their own antiparticles. BST says: it happens, but so faintly (1.5–3.7 meV) that even these detectors won't see it. So BST makes a clean forward call — *they will not detect it* — and a kill switch. The honest kill switch is the brightness the *inverted* mass-ordering would give, around 15 meV and up: a detection there means the lightest neutrino is *not* massless, and BST is done — that one holds no matter how the input measurements shake out. A glow in the awkward middle (roughly 4 to 15) would be real tension worth staring at, but the clean, no-wiggle kill is the bright one. The

current best limit (KamLAND-Zen, now peer-reviewed, 2025) sits far above even that, so a “nothing yet” can’t touch us; only a detection can, and only in the wrong place.

- The ordering (JUNO). A reactor experiment in China will nail down whether the ladder climbs “normal” (our prediction) or “inverted,” around 2030. Inverted kills us.
- The octant and the ordering again (DUNE, in the US). Settles the upper-vs-lower angle and sharpens the ordering. A decisive *lower* octant kills us.

Three experiments, three ways to be wrong, all this decade, all against a theory with no dials. Pick the one you trust least and watch it.

What this is not

- It is not “we derived the neutrino masses from nothing.” We derive the *zero*, the *ordering*, the *angles’ structure*, and that CP is *violated at all* — but not the value of the CP phase itself (that one, BST predicts only that it exists, not what it is — we mark it as a prediction to be measured, not a derived number). The overall *scale* — how heavy the two massive ones actually are — still leans on the two measured mass-differences. We take those as given, plainly, the way every theory takes some measured anchor. We mark exactly where we lean.
- It is not “every fraction here is forced.” The individual mixing-angle fractions are consistent and pretty, but a small fraction near a measured number is cheap, and we hold them as consistent, not proven. The claims we *stake* the theory on are the ones with teeth: the exactly-massless neutrino, the sub-4-meV glow, the upper octant, the dark-energy weld.
- It is not a story where the 58 meV total is a BST triumph. That number is the generic floor of any normal-order theory. Our distinctive content is the *shape* around it, not the number.

The one shape

Strip everything away and here’s the whole theory in a sentence: the geometry has room for two, and there are three, so one is left over — and that leftover is a massless neutrino, an unpaired ground state, and (it turns out) the reason a rare decay glows exactly as dim as it does. One asymmetry, read in three places. That’s not a coincidence we arranged; it’s the same rank-two fact, and it’s why the predictions are *coupled* — you can’t move one without the others, which is exactly what makes the theory easy to kill and worth testing.

The math is public. The two mass-differences we use are on the record; every prediction has a short program next to it that you can run; and three experiments will render a verdict within about a decade. Pick the row you trust least and check that one. If a single row fails, you’ve learned something real — and so have we.

The lightest neutrino weighs nothing. Break that if you can.

Notes for the team (not for the reader): NARRATIVE pilot, drafted by Keeper in the Guide voice, companion to the technical T2563 + Grace’s Falsifier Data-Sheet. Holds the tier discipline: distinctive ($m_1=0$ -exact, sub-4-meV $0\nu\beta\beta$, θ_{23} upper octant, the DE weld) separated from generic ($\Sigma m \approx 58$ = minimal-NO floor) and consistent-not-forced (the mixing-angle fractions, per K1464); the absolute scale leans on the two measured splittings (identified, stated plainly); the retired m_3/m_2 ratio not reclaimed; δ_{CP} existence-only (not narrated as a match). The “breaking is the content” / one-ground-state=one-massless-neutrino through-line is the spine (T2563 Section 8). Owes: Cal’s independent tier-inflation re-read of THIS voice (the warm register is where a tier quietly inflates — same as the Axioms pilot); pairing with the academic/technical version + Grace’s data-sheet; Casey GO. Nothing external without both voices + Cal-vet + Casey GO.